

(30 marks in total) Consider the linear differential equation

$$\ddot{y} + 5\dot{y} + 4y = \dot{u} + 3u$$

with initial condition $y(0) = 1$ and $\dot{y}(0) = -1$. Let $S = (A, B, C, D)$ be the second order control canonical representation for the system. Answer the following questions:

- (a) (15 marks) Compute e^{At} .
- (b) (5 marks) If the input is such that $u(0) = 0$, determine the initial condition for the control canonical form corresponding to the given initial state for the differential equation.
- (c) (10 marks) Solve the linear differential equation for the given initial conditions by solving the state-space equation for $y(t)$ for any input u with $u(0) = 0$.

(35 marks in total)

- (a) Consider the nonlinear ODE

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \sigma(y - x) \\ rx - y - xz \\ -bz + xy \end{bmatrix},$$

with $\sigma, b > 0$ and $0 < r < 1$. Answer the following questions:

- (i) (5 marks) Show that the ODE has only a single equilibrium point at the origin.
- (ii) (7 marks) Consider the candidate Lyapunov function

$$V(x, y, z) = x^2 + \sigma y^2 + \sigma z^2.$$

Show that $\dot{V}$ is a quadratic form on $\mathbb{R}^3$.

- (iii) (8 marks) Using the candidate Lyapunov function from sub-problem (a-ii), show that the origin is globally asymptotically stable.
- (b) (15 marks) A rotating rigid satellite with state $\omega = (\omega_1, \omega_2, \omega_3)^T$ has dynamics given by:

$$\begin{aligned} \dot{\omega}_1 &= \frac{J_2 - J_3}{J_1} \omega_2 \omega_3 + \frac{u_1}{J_1} \\ \dot{\omega}_2 &= \frac{J_3 - J_1}{J_2} \omega_3 \omega_1 + \frac{u_2}{J_2} \\ \dot{\omega}_3 &= \frac{J_1 - J_2}{J_3} \omega_1 \omega_2 + \frac{u_3}{J_3}, \end{aligned}$$

where ω_1 to ω_3 are the angular velocities along the satellite's principal axes, u_1 to u_3 are the control signals that drive the torques applied about the principal axes, and $J_1, J_2, J_3 > 0$ are the principal moments of inertia. Consider the candidate Lyapunov function

$$V(\omega) = \omega^T J \omega,$$

with $J = \text{diag}(J_1, J_2, J_3)$. Let $u = (u_1, u_2, u_3)^T$ and consider a linear state feedback law $u = -K\omega$, with $K \in \mathbb{R}^{3 \times 3}$. How would you choose K so that the control law renders the origin globally asymptotically stable? Prove your claim.

(35 marks in total) A mobile robot is moving along a 2D plane with axes X and Y . At any time, the robot will have a position (x, y) on the plane and is moving at some velocity v in the direction of a heading angle θ with respect to the X -axis, see Fig. 1. You are able to remotely control the robot's acceleration along its heading direction and the rate of change of its heading. The state of the robot is given by $\xi = (x, y, v, \theta)^T$ and the motion of the robot is given by the controlled nonlinear ODE:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{v} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v \cos(\theta) \\ v \sin(\theta) \\ u_1 \\ u_2 \end{bmatrix}.$$

In the above, u_1 is a control signal for the acceleration along the robot's heading and u_2 is a control signal for the rate of change of heading.

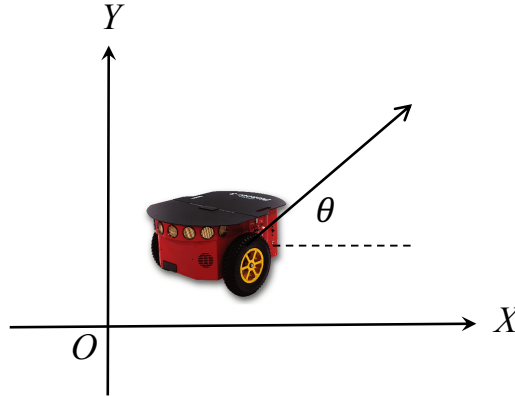


Figure 1: Robot motion on a 2D plane.

Let (x_T, y_T) be some fixed position on the 2D plane and let θ_T be some fixed orientation in $(-\pi, \pi]$. Consider the candidate Lyapunov function

$$V(\xi) = \frac{1}{2} ((x - x_T)^2 + (y - y_T)^2 + v^2 + (\theta - \theta_T)^2).$$

Answer the following questions:

- (a) (10 marks) Compute $\dot{V}(\xi)$.
- (b) (5 marks) Determine state-feedback control signals u_1 and u_2 such that $\dot{V}(\xi) = -\kappa v^2 - \gamma(\theta - \theta_T)^2$ for some constants $\kappa, \gamma > 0$.
- (c) (15 marks) Under the control signals in sub-problem (b), let $\Omega_{r,c} = \{\xi \in \mathbb{R}^4 \mid \|\xi - (x_T, y_T, 0, \theta_T)^T\| \leq r, V(\xi) \leq c\}$ be some positively invariant set under the robot's dynamics for some $r, c > 0$. Describe the largest positively invariant set M in the set $I = \{\xi \in \Omega_{r,c} \mid \dot{V}(\xi) = 0\}$. That is, $M = \{\xi \in I \mid \dots\}$, fill in the
- (d) (5 marks) Suppose that $(x_T, y_T) = (1, 1)$ and $\theta_T = \pi/4$ radians. Make a sketch of the possible final positions (x, y) that the robot could end up at on the plane (as $t \rightarrow \infty$) under the control law in sub-problem (b).